

Food*Lens*

Transparent multi-objective food recommendations via contrastive XAI

Health + Eco — shown together, never averaged, always explained.

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Outline

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- 2 **XAI methods selection**
- 3 **Key insights from user testing**
- 4 **Technical & design challenges**
- 5 **Live demonstration of the dashboard**
- 6 **Lessons learned & future improvements**

Section 1

Design process & *rationale*

1

WA3 discovery: research before code

We ran **six semi-structured interviews** with people who make real grocery decisions — from a 20-year-old student to a 41-year-old designer on a quasi-vegetarian diet.

- Protocol kept **context exploration separate from XAI concepts** to avoid leading questions
- Thematic analysis (Braun & Clarke): **47 codes -> 5 Key Insights**
- The asymmetry we found: **health is a label you read in passing; ecology is buried in a database** — and the conflicts between them are surfaced nowhere

Every feature we shipped traces back to one of those insights. Nothing was designed by intuition alone.

The five Key Insights

- **KI-1 — Visible reasoning is a prerequisite for trust.** An opaque score reads as marketing. Never hide the "why".
- **KI-2 — The value-action gap is contextual.** Time, budget, fatigue override good intentions. Never moralise.
- **KI-3 — Single-axis labels create axis blindness.** A soda outscoring juice; water-hungry almonds looking "healthy".
- **KI-4 — Contrastive framing turns explanations into decisions.** "30% less sugar than yours" beats "70/100".
- **KI-5 — Tolerance for configuration varies widely.** Sensible default, presets, advanced slider behind progressive disclosure.

From insights to seven Design Hooks

Hook	Rule it enforces
H1	Dual-axis default — eco badge never hidden, even "no data" renders with a caption
H2	Reasoning IS the recommendation — contrastive sentence open by default
H3	One sentence, one number — charts are drill-downs, not the entry point
H4	Configurable first axis — health/eco slider; price behind "Advanced"
H5	Compare-this-shelf — N products in, ranked list out, one-line justification each
H6	Minimal onboarding — goals + age + gender only
H7	Close into action — every card ends in a real next step

Personas: who we are designing for

- **Marc Vidal** — time-strapped pragmatist. Shops on autopilot, will not configure. Wants instantly visible reasoning.
- **Pau Estarellas** — sceptical analyst. Distrusts opaque scores. Needs verifiable numbers and the drill-down.

- **Lluís Tomàs** — autonomous planner. Hates greenwashing and moralising. Wants quick visual graphs, opt-in only.
- **Aina Servera** — convenience seeker. Time- and budget-bound. Hard boundary against guilt-inducing UX.

XAI methods selection

Why these methods: Miller's criteria

Tim Miller (2019): good explanations are **contrastive, selective, and grounded in the user's mental model** — not raw scores.

We chose each XAI component to satisfy a specific criterion:

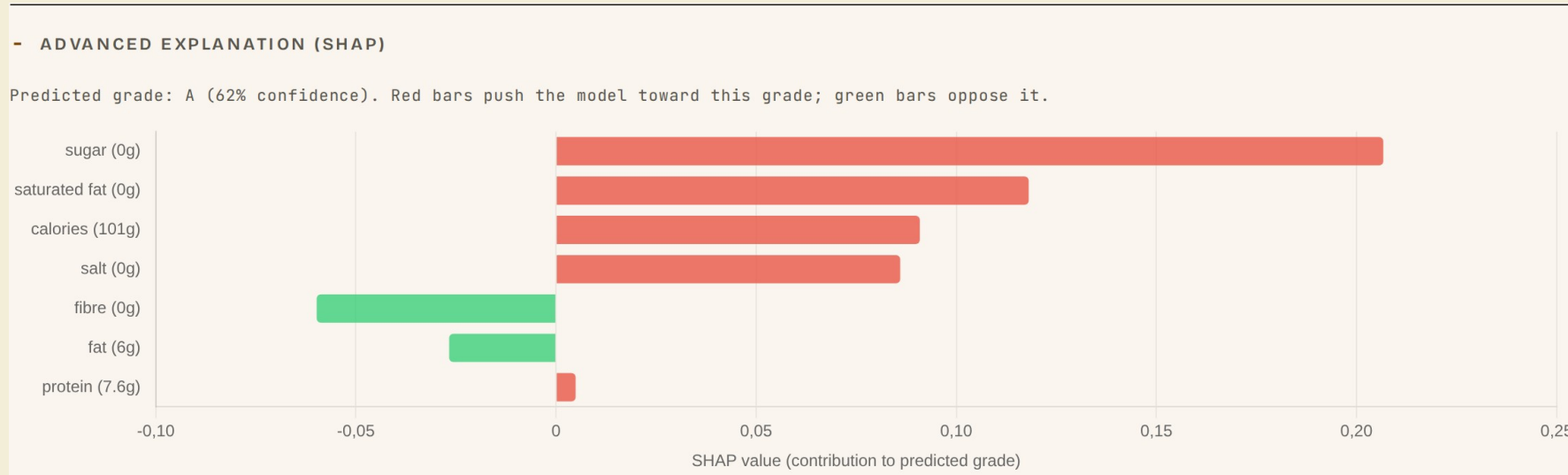
- **Contrastive** -> "why this and not your usual?" — the contrastive sentence generator
- **Selective** -> one number, not twenty — H3 discipline over rich model output
- **Grounded / auditable** -> an interpretable model plus a verifiable drill-down for the sceptic

The literature gave us the test; our job was to make a vanilla-JS app pass it.

An interpretable model: Random Forest for Nutri-Score

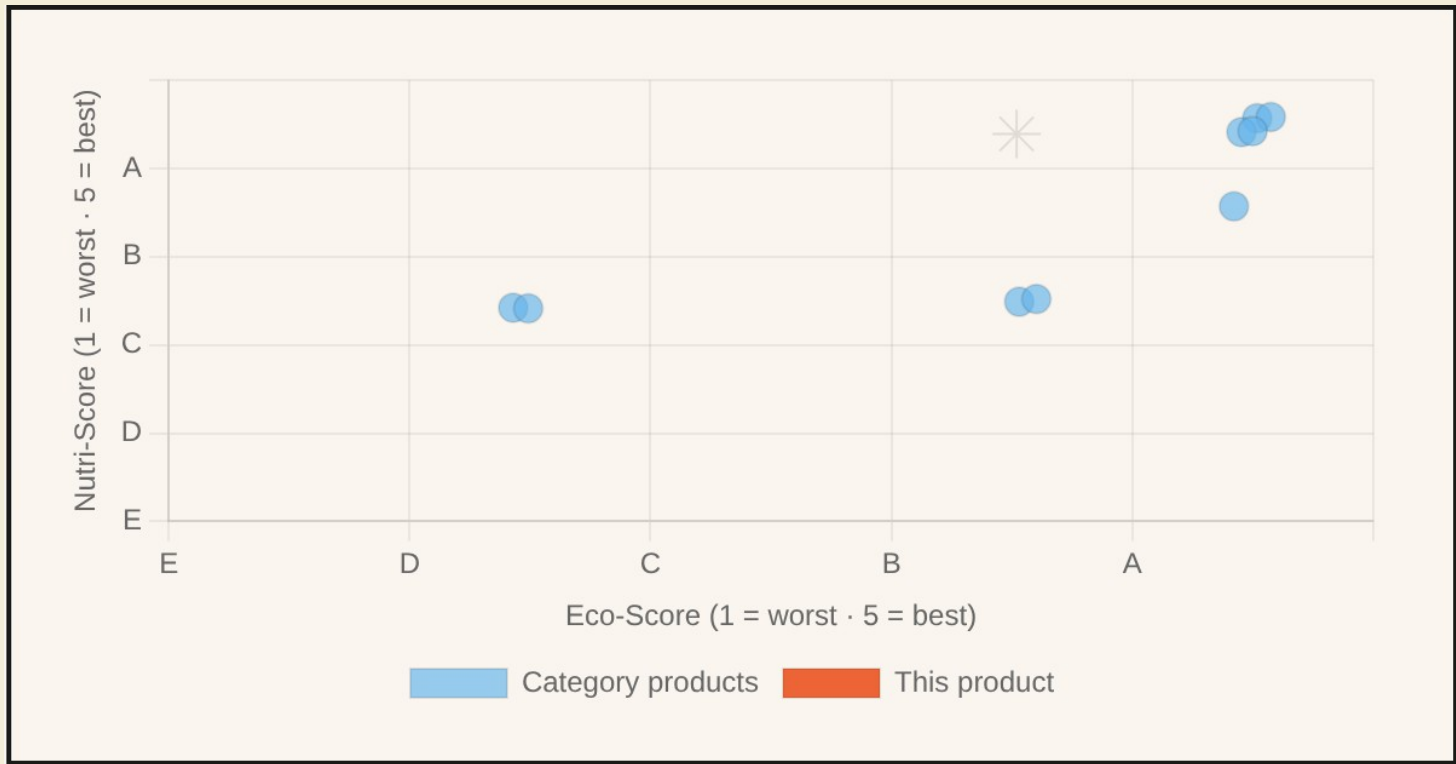
- The system reads the **official `nutriscore\_grade`** — we never recompute the certified label locally.
- For the XAI pipeline we trained a **Random Forest** on a real OFF dump (~80% five-fold CV accuracy).
- Why a forest, not a deep net: **feature importances are first-class** — fat, sugars, saturated fat, salt, protein, fibre, energy each carry a readable weight.
- This is a **pipeline-verification model**: it proves the explanation path end-to-end, it does not replace the official grade.

SHAP -> one contrastive sentence, one number



- **SHAP local attribution** computes each nutrient's contribution for *this* product
- We compress that vector into **one contrastive sentence with exactly one verifiable number** (H3)
- The full SHAP detail is parked behind an "**Advanced explanation**" — selective by default, complete on demand

The scatter drill-down: for the sceptic who verifies



- The contrastive sentence answers Marc in half a second; **Pau Estarellas wants to see the space.**
- `GET /api/scatter` returns a **visual XAI scatter plot**: where this product sits on the health/eco plane relative to its category.
- The drill-down is **opt-in** — Miller's selectivity, and Lluís's "no moralising, just show me the graph".

How each method helps the user: sentence = instant trust; SHAP advanced view = traceability; scatter = independent verification.

The AI we used — in the product and in development

In the product (ML / XAI)

- **Random Forest** (scikit-learn) — predicts Nutri-Score for the explanation pipeline
- **SHAP** — local feature attribution -> the one contrastive sentence
- **Weighted KNN** — the Alternative Engine (healthier / greener substitutes)
- **DuckDB** — queries the full OFF dump at scale

In development (generative AI)

- **LLM coding assistants** to speed up boilerplate, tests, and docs
- **Human-verified**: every change reviewed, adversarially audited (this caught the dead allergen filter), and gated by tests
- We **direct** the AI — the design and the verification are ours

Key insights from user testing

Two instruments, built in-app (F-46)

Evaluate FoodLens

Use these WA5 instruments after a participant completes the search, comparison or barcode flow.

System Usability Scale

1 means strongly disagree. 5 means strongly agree.

1. I think that I would like to use FoodLens frequently.

1

2

3

4

5

2. I found FoodLens unnecessarily complex.

1

2

3

4

5

3. I thought FoodLens was easy to use.

1

2

3

4

5

4. I think that I would need support to use FoodLens.

1

2

3

4

5

5. I found the different FoodLens functions well integrated.

1

2

3

4

5

6. I thought there was too much inconsistency in FoodLens.

1

2

3

4

5

7. I would imagine that most people would learn to use FoodLens quickly.

1

2

3

4

5

8. I found FoodLens cumbersome to use.

1

2

3

4

5

9. I felt confident using FoodLens.

1

2

3

4

5

10. I needed to learn a lot before I could use FoodLens.

1

2

3

4

5

- SUS (Brooke 1996) — 10 items, usability of the dashboard (KI-2)
- Explanation Satisfaction — anchored to YEO (Wijkman 2024): ¹clarity, consistency, on-demand detail, reliability/fairness

Procedure & cohort

- **Six task scenarios** mapped to documented user flows: onboarding+search · read explanation · find alternative · compare shelf · set "my usual" · drag slider
- **SEQ** after each task for per-flow diagnostics; SUS once at the end
- **Cohort:** the six WA3 interviewees re-used (Nielsen formative, n=6). P01 & P03 are team members — declared bias, weighted as optimistic
- Scoring **pre-registered:** odd items raw minus 1, even items 5 minus raw, then x2.5 to 0-100; benchmark **68**, Sauro–Lewis grade bands

Honest status: instrument built, data pending

- The instrument is **built, wired, and ready**. The task suite and cohort are defined.
- **No participant data has been collected yet**. We will **not** quote a SUS score we did not measure.
- Pre-committing the scoring rules and thresholds **before** collection is sound method — it removes the temptation to fit interpretation to whatever numbers arrive.
- **Pre-registered hypotheses:** (H-a) contrastive + drill-down raises satisfaction for the sceptic; (H-b) zero-config default -> high SEQ for the pragmatist; (H-c) any moralising copy erodes trust for the convenience seeker.

Section 4

Technical & *design challenges*

4

Open Food Facts: the API fought back

- **Intermittent 503s** on `/search` -> silent fallback to `sample_products.json`, non-blocking banner keeps the UI alive
- **Missing `User-Agent`** -> **403**. Mandatory header `FoodLens-MVP/0.1` on every call
- **Field rename**: read `environmental_score_grade` **first**, fall back to legacy `ecoscore_grade`
- **Rate limits** (search 10/min) -> **Redis** cache budgeting on the backend
- **Graceful degradation**: backend times out at **1500 ms** (KNN) / **2000 ms** (explanation), then falls back to the in-browser path

Design discipline under pressure: keep the **eco badge visible** for "not-applicable" (H1); hold **one-sentence-one-number** even when SHAP hands us a rich vector (H3).

Honest QA: defects we found and fixed before release

An adversarial review + engineering pass caught **three real defects** — none of which a usability test would have caught:

- **Allergen filters were dead.** `api.js` never requested `allergens_tags`, so cards carried no allergen data — a **safety issue**. Fix: request the field, normalise to tokens, exact OFF-token matching.
- **Category browser search was commented out.** `runSearch` wiring was dead. Fix: connected dropdown filters to live in-memory rendering.
- **Falsy-zero bug.** A genuine **0 g** rendered as "no data" under a truthiness test. Fix: explicit `typeof`/null guards — real 0 shows "0"; missing shows "no data".

We disclose these candidly. Finding them strengthens the build; hiding them would not.

Section 5

Live demonstration

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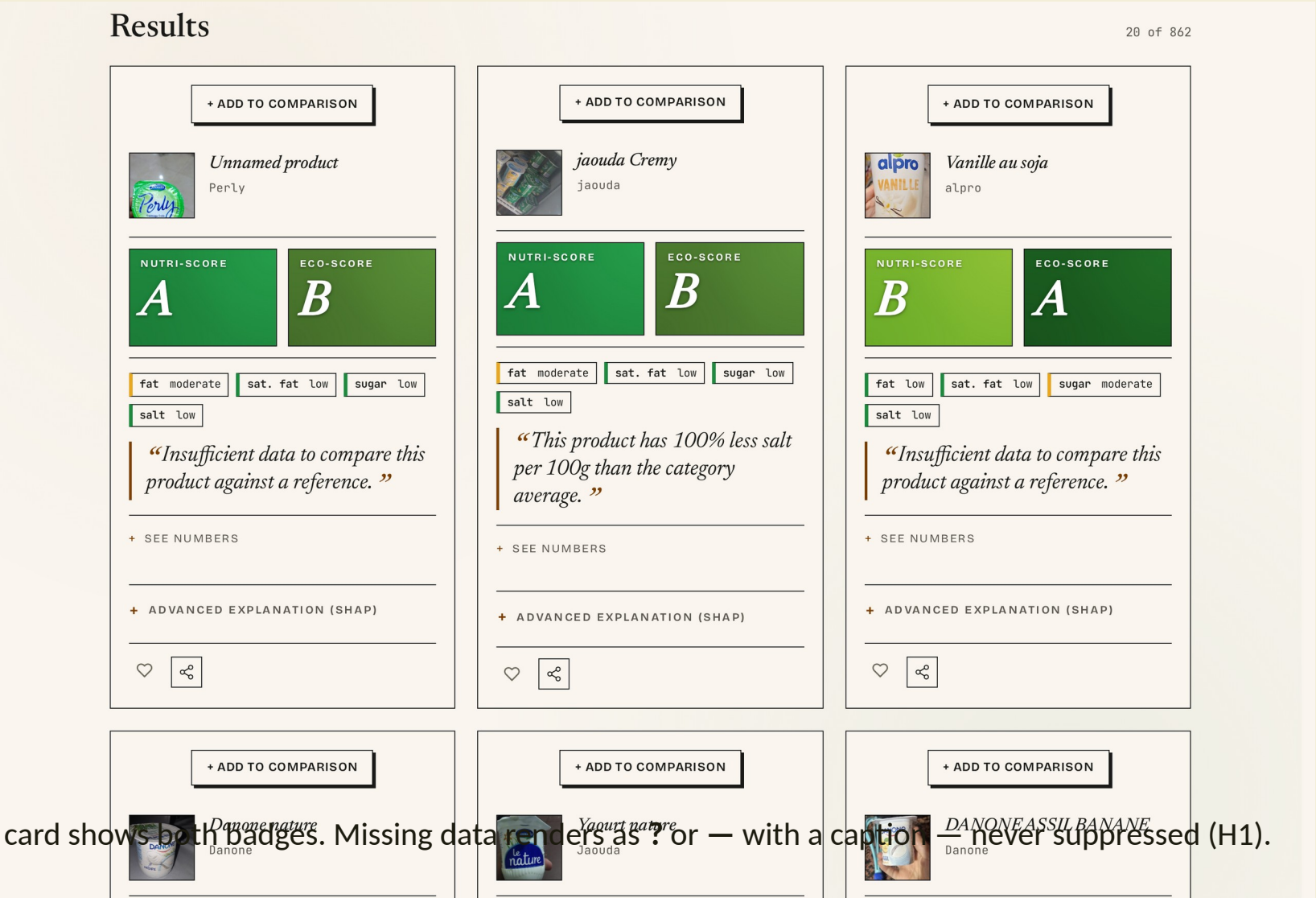
What we will show live (about 8 min)

1. **Onboarding** — goals + age + gender only (H6), then a product search
2. **Dual-axis card** — Nutri-Score and Eco-Score side by side, contrastive sentence **open by default** (H1, H2, H3)
3. **Advanced explanation** — expand SHAP attribution; open the scatter drill-down
4. **Priority slider** — drag health/eco; show the ranking actually changes (H4, KI-5)
5. **Filters** — allergen + eco multi-criteria filtering; category browser (Batch-I)
6. **Compare this shelf** — N products -> ranked list + winner-highlight table -> **CSV export** (H5)
7. **i18n** EN/ES/CA · barcode scan · dark mode · PWA

AI live, not mock-ups: the SHAP explanation + scatter (step 3) and the KNN Alternative Engine (steps 5–6) run on the live scikit-learn + SHAP backend.

Fallback: if the live app fails, the next two slides are screenshot backups.

Demo backup A — dashboard & dual-axis grid



Demo backup B — comparison view

Compare your shelf

A ranked list out, one verifiable line per product. Add items from Discover, then read them side by side.

BACK TO DISCOVER

EXPORT CSV

Ranked for your health / eco priority

1. 6111242100985

Best overall: Nutri-Score A, Eco-Score B, 100% less sugar than the group average.

2. jaouda Cremy

Nutri-Score A, Eco-Score B, 37% less sugar than the group average.

3. Vanille au soja

Nutri-Score B, Eco-Score A, 137% more sugar than the group average.

Attributes	6111242100985	jaouda Cremy	Vanille au soja
Nutri-Score	A	A	B
Eco-Score	B	B	A
Energy (kcal)	101	109	65 ★
Sugars (g)	0.0 ★	2.0	7.5
Fat (g)	6.0	3.0	2.2 ★
Saturated Fat (g)	0.0 ★	0.4	0.4

Ranked summary + one-line contrastive justification per product + attribute table highlighting the winner per axis. Genuine **0 g** renders as "0", not "no data".

Lessons learned & *future improvements*

Lessons learned

- **Research pays for itself.** Five Key Insights gave us a non-negotiable spec — we never argued about features from taste.
- **Honesty is a feature.** Disclosing dead filters and pending data is more credible than a polished fiction. Missing is not zero is a discipline, not a detail.
- **Constraints enabled the work.** No build step, no framework — a vanilla app runs with a double-click, and graceful degradation kept it alive when the API didn't cooperate.
- **XAI is an interaction problem, not a model problem.** The hard part was compressing SHAP into one trustworthy sentence, not training the forest.

Next Steps

Future work & close

- **Run the study** and populate the SUS / XEQ tables with real figures
- **SHAP over DuckDB** in the backend; scale the index from ~862 toward the full OFF catalogue
- **Price as a first-class axis** behind Advanced (H4) — for Aina
- **Playwright** regression suite; longitudinal in-the-wild study vs. a Yuka baseline

Food*Lens*

Dual-axis · Contrastive · Honest — github.com/AlejandroB10/foodlens

Thank you — questions?